MILITARY SPECIFICATION

CHLORZOXAZONE AND ACETAMINOPHEN TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Chlorzoxazone and Acetaminophen Tablets (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

Federal

FED-STD-140	Tablets (for Medicinal Purposes).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Chlorzoxazone and Acetaminophen Tablets and shall be in accordance with all applicable requirements of FED-STD-140 and as specified herein:

S2. Classification. Shall be type I.

S.5.3 The following is added to this paragraph:

Shall be tablets containing 250 mg chlorzoxazone and 300 mg acetaminophen within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 92.5 percent and not more than 110.0 percent of the required amounts of chlorzoxazone and acetaminophen when determined as specified in 4.3.1.

The chlorzoxazone and acetaminophen shall be identified by the typical shape of the spectral curve as determined in the assay.

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S6.4.9.1 The tablets shall disintegrate in not more than 30 minutes when tested according to the USP Disintegration Test for uncoated tablets.

3.2 Ingredients.

3.2.1 Chlorzoxazone. The chlorzoxazone used in the manufacture of the tablets shall comply with the following:

3.2.1.1 Melting range. Shall be not less than 189° C, and not more than 194° C, when determined by the U.S.P. method for class I.

3.2.1.2 Identification. Determine the absorbance, using a suitable spectrophotometer, of a solution containing 2 mg chlorzoxazone in 100 ml of methanol in the spectral region between 220 nm and 300 nm using 1 cm cells. The solution shall exhibit a minimum (less than or equal to 0.08) at 248 ± 2 nm and a maximum at 282 ± 1 nm with inflections at about 225 nm and 288 nm.

3.2.1.3 Limit of 2-amino-4-chlorophenol. Shall exhibit no more orange color than a standard of 2-amino-4-chlorophenol when determined as specified in 4.3.2.1.

3.2.1.4 Limit of cyanide. Shall exhibit no more orange color than a standard of potassium cyanide equivalent to 4 mcg of cyanide (20 ppm) when determined as specified in 4.3.2.2.

3.2.1.5 Loss on drying. Shall be not more than 0.5 percent when a suitable sample is dried at 105° C for 2 hours, cooled to room temperature and weighed.

3.2.1.6 Residue on ignition. Shall yield not more than 0.15 percent sulfated residue on ignition when determined by the U.S.P. method for Residue on Ignition.

3.2.1.7 Chlorine content. Shall be not less than 20.6 percent and not more than 21.2 percent on a dry basis when determined as specified in 4.3.2.3.

3.2.1.8 Heavy metals. Shall be not more than 20 ppm using a solution in acetone (reference U.S.P. aspirin). Correct by color readings for color of compound. NOTE: Double the quantity of standard.

3.2.1.9 Assay. Shall assay to contain not less than 97.0 percent and not more than 103.0 percent chlorzoxazone when determined as specified in 4.3.2.4.

3.2.2 Acetaminophen. The acetaminophen used in the manufacture of the tablets shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

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3.2.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1.

3.3 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling plan as shown in FED-STD-140 shall be applicable.

4.2.1 Samples. If (i) the quality assurance representative submits samples to the laboratory at the Defense Personnel Support Center, or (ii) the bidder (offeror) is required, by the terms of the Schedule or otherwise, to submit samples to said laboratory, then a quantity of each lot of chlorzoxazone and of acetaminophen used in the manufacture of each lot of tablets, sufficient to determine all of the required tests, and 0.1 gram of chlorzoxazone standard accompanied by a certificate of analysis listing all test results, shall be submitted with the finished product to the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, Attention: Quality Assurance Branch, 2800 South 20th Street, Philadelphia, PA 19101.

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4.3 Tests.

4.3.1 Assay for chlorzoxazone and acetaminophen.

Standard. Prepare a standard containing 250 mg chlorzoxazone and 300 mg acetaminophen diluted so that the final concentration of chlorzoxazone is 0.5 mg per 100 ml.

Procedure. Weigh and finely powder not less than 20 chlorzoxazone and acetaminophen tablets. Weigh accurately a portion of the powder equivalent to 25 mg chlorzoxazone and transfer it to a 100-ml volumetric flask. Shake for 15 minutes with 80 ml of warm methanol. Add methanol to volume, and mix.

Filter the solution through a dry filter paper, rejecting the first 20 ml. Transfer a 2 ml aliquot to a 100-ml volumetric flask. Add methanol to volume, and mix.

Determine the absorbance of the clear sample and standard in 1-cm quartz cells at 247 nm and at 282 nm with a suitable spectrophotometer using methanol as the blank.

Calculate the percent of the required amounts of chlorzoxazone and of acetaminophen as follows. The readings from the tablet excipients are negligible.

% of chlorzoxazone = $A_t1/A_s1 \times 100$

% of acetaminophen = $A_t2/A_s2 \times 100$

Where A_t1 and A_t2 are the absorbances of the solution representing the tablets at 282 nm and 247 nm, respectively, and A_s1 and A_s2 are the absorbances of the standard mixture at 282 nm and 247 nm, respectively.

4.3.2 Chlorzoxazone (ingredient).

4.3.2.1 Limit of 2-amino-4-chlorophenol. Dissolve 50 mg of chlorzoxazone in 25 ml of methanol. Prepare a standard by dissolving 0.125 mg of 2-amino-4-chlorophenol in 25 ml of methanol. To the sample and standard add 1 drop of 1 percent (w/v) ferric chloride solution and 1 drop of 5N acetic acid. Five minutes after the addition of the reagents compare the colors.

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4.3.2.2 Limit of Cyanide.

Place 2 grams of chlorzoxazone in the flask of a semimicro steam distillation apparatus. Add 15 ml of purified water. Add 1 gram of tartaric acid powder and quickly place into position.

Steam distill 5 ml of distillate into a test tube containing 1 ml of 1 N sodium hydroxide and 3 ml of purified water. Make acid to litmus with diluted hydrochloric acid (0.5 ml) and immediately add 0.5 ml 10 percent (w/v) solution sodium carbonate anhydrous. Add 0.4 ml of a saturated aqueous solution of picric acid.

Immerse in boiling water for 5 minutes. Cool to room temperature.

After 5 minutes at room temperature compare the sample to the standard for color.

4.3.2.3 Chlorine content. Weigh accurately about 300 mg of chlorzoxazone powder. Transfer to a 250 ml dry reflux flask. Dissolve in 10 ml U.S.P. ethanol and add 3.5 grams Raney nickel-aluminum catalyst. Place under a reflux condenser, chill flask in an ice bath and add through the condenser 75 ml of 10 percent w/v sodium hydroxide solution. After the initial reaction has subsided, remove ice bath. In 5 or 10 minutes begin heating the reaction mixture gently, increasing the heat gradually to maintain a fairly rapid reflux. Remove the heat 1-1/2 hours after the addition of the alkali. Cool and rinse the condenser. Decant the liquid into a 200-ml volumetric flask. Wash the nickel residue with purified water by decantation and dilute the mixture to exactly 200 ml. Transfer exactly 100 ml to a 400 ml beaker. Cool, neutralize and then acidify to congo red paper by adding about 13 ml of concentrated nitric acid dropwise with constant swirling. Determine the chloride by titrating with 0.1N silver nitrate. The equivalence point is detected potentiometrically using silver and calomel electrodes. Each ml of 0.1N $AgNO_3$ is equivalent to 3.546 mg of chlorine.

4.3.2.4 Assay. Determine the abosrbance of a solution containing 2 mg of chlorzoxazone in 100 ml of methanol at 282 nm, using 1 cm cells and methanol as a blank.

Calculation: 100 $(A_{282})/0.635$ = percent chlorzoxazone.

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4.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing five hundred (500) tablets as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Twenty-four (24) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the national stock number.

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Notes: (1) The national stock number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the national stock number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I Shelf life markings shall be shown as specified in MIL-STD-129.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking shall be shown as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-764-3313	CHLORZOXAZONE AND ACETAMINOPHEN TABLETS, 500s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1451
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37304 CHLORZOXAZONE AND ACETAMINOPHEN TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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